CHETAN SINGHAL| ME22B038 | INDIAN INSTITUTE OF TECHNOLOGY MADRAS

EDUCATION AND SCHOLASTIC ACHIEVEMENTS

Program	Institute	% / CGPA	Year
B.Tech in Mechanical Engineering	Indian Institute of Technology, Madras	8.9/10	2026
Class XII (RBSE)	Happy Convent School	91.4%	2022
Class X (RBSE)	Happy Convent School	95.8%	2020

- Achieved **AIR 5128** in **JEE Advanced'22** and top **1%** (JEE Mains'22) among **1.15M+** candidates, showcasing **problem-solving, analytical skills**
- Secured Rank **13** (**Rajasthan**) and International Rank **259** in the prestigious **International Math Wizard Olympiad** held globally back in **2015**
- Selected amongst **1L+** qualified applicants, earning a **₹2 Lakh** merit-based **scholarship** in the highly competitive **Reliance Foundation Exam**
- Achieved a **Leetcode rating of 1690** & successfully solved **300+ DSA[1] problems** across multiple competitive coding platforms (GFG, Atcoder)

COURSES & CERTIFICATIONS

Data Structures & Algorithms	Machine Learning	Deep Learning	Probability (MA2040) *	Differential Equations
MLF (Linear Algebra) *	Function in Several Variable	Climate Economics	Series & Matrices	Structured Query Language

Programming	C++, Python, C, SQL[2] (PostgreSQL, MySQL), JavaScript, MATLAB, HTML, Tailwind CSS, DSA, OOPs[3], REST/Fast APIs, pytorch
Software Tools	PyCharm, VS Code, Jupyter Notebook, Git, GitHub, Docker, Postman, Render, CloudflareR2, Tableau/Power BI, Figma, Aws
ML[4]/DL[5]/NLP[6]	Supervised Learning, Unsupervised Learning, Gen AI, RNN, CNN, LSTM, NLP, RL, CV[7], Autoencoder, Transformer, Langchain

PROFESSIONAL EXPERIENCE

NOKIA
Data Science Intern
(Jan'25 - May'25)

Automated telecom data workflows with parsing, translation, integration, Excel DSL, and ML, improving efficiency 20%.

- Made a **PyShark parser** for 100+ PCAP8 files, pulling **500K+ packets** and boosting analysis 5×, cutting Wireshark use **95%**
- Engineered a pipeline with **lxml & Google Deep Translator** for **100+ DOCX files, cutting effort 90%** via batch translation
- Designed an engine merging **lncel/nrcell/nrfim dataset**, automating reports for **200K+ 5G cells & cutting QA effort 85%**
- Created an **OpenAI DSL-powered Excel engine,** enabling row **comparison** with thresholds and **cutting audit cycles 90%**
- Applied **ML on telecom data, boosting performance 15%** by identifying bottlenecks and **optimizing overall traffic flow**

TAFE
Software Development Intern
(May' 25 -Jul'25)

Built an inventory management platform with automation and real-time dashboards, boosting productivity by 80%.

- Boosted overall productivity **80%** with a **full-stack inventory system** using **Flask, PostgreSQL, Render, and CloudflareR2.**
- **Reduced fraud** and manual tracking **95%** with **geo-tagged attendance & live snapshots**, significantly enhancing accuracy
- **Saved ₹4.5L** annually by **automating usage tracking & Inventory leakage** prevention through smart monitoring systems
- Increased overall productivity **80%** with real-time role-based dashboards and **automated SMTP notifications** integration
- Ensured **accountability 100%** by securing dashboard access with **Flask-Login, PostgreSQL, & JWT-based authentication**

PERSONAL PROJECTS

Vision OCR

Using LLaMA with LoRA fine-tuning, Unsloth, 4-bit quantization, multi-metric evaluation, and automated preprocessing

- Developed OCR[12] pipeline using **Llama-3.2-11B-Vision**, achieving **91% reduction** in Word Error Rate through **fine-tuning**
- Implemented **LoRA fine-tuning** with Unsloth and **4-bit quantization**, improving **BLEU[13] score by 156% and CER by 90 %**
- Engineered **multi-metric evaluation framework** with an **automated preprocessing pipeline** for production-ready OCR

DocuChat AI

Built Streamlit-based multi-PDF ChatApp using LangChain, FAISS, Google GenAI, OpenAI GPT-3, Claude, and LLaMA

- Upload multiple PDFs/TXT files, process with **LangChain chunking and enable multi-document** conversational querying.
- Integrates **FAISS vector database seamlessly** with **LLMs like Gemini Pro, GPT-3, Claude & LLaMA2** for efficient retrieval.
- Built using **Streamlit UI, LangChain pipelines**, and **Google GenAI SDK for embeddings** and seamless model interaction.

Crop Disease Detection

EfficientNetB3-based TensorFlow model for plant disease detection, trained on 87K images, achieving 98.47% accuracy

- Developed a deep learning model in **TensorFlow** using **EfficientNetB3** to classify diverse **plant leaf diseases across crops**
- Preprocessed and augmented **60k images** with flips, rotations, zoom and shifts to enhance **robustness & generalization**
- Achieved **99.13% training and 98.47% validation accuracy**, demonstrating reliability for **agricultural disease diagnosis**

Twitter Sentiment Analysis

Implemented an advanced NLP pipeline leveraging Tokenization, TF-IDF[14], Word2Vec, and LSTM for sentiment analysis

- Converted raw text into structured features via **Tokenization** and **TF-IDF** for highly efficient & effective model readiness
- Enhanced text representation using **Word2Vec embeddings** integrated with **NLTK[15]** for a **richer semantic** understanding
- Built **LSTM[16]** - based sequential model achieving **0.91 F1 score**, validated with a **confusion matrix & classification report**

Pixel Classification

U-Net architecture with skip connections, TensorFlow/Keras, IoU & Dice on the CARLA semantic segmentation dataset

- Developed **U-Net** semantic segmentation model in **TensorFlow/Keras on CARLA** dataset for pixel-wise object detection
- Built a **preprocessing pipeline** to **load, resize, normalize, & batch** image–masks pairs, enabling efficient model training
- Achieved **95.4% pixel accuracy**, validated using **IoU[11] & Dice metrics coefficient** for fine object boundary segmentation

POSITIONS OF RESPONSIBILITY

Heritage Club

Head *(May'24 – Mar'25)*

- Successfully organized **13+** major events for **11.5K** students, engaging nostalgia tours for about **2.5K+ alumni** nationwide
- Increased overall annual Centre footfall by 300% via **heritage film, VR exhibitions, ACK comic** and magazine relaunches

Manager *(June'23 – Mar'24)*

- Spearheaded immersive exhibitions, **alumni nostalgia tours, & heritage publications** to significantly boost engagement
- Led heritage tours for freshers to **Arikamedu, Mahabalipuram, and regional museums**, promoting cultural exploration

CDC

Dep. Head *(May'24 – Mar'25)*

- Led career development initiatives for **13K+ students**, **managing DAMP[18], 15+ workshop**s, webinars, high-impact events
- Organized **36+ hour hackathon, coordinated national-level mock exams**, created HS bluebook for **10+ top universities**

Manager *(June'23-Mar'24)*

- Assisted in creating the CDC Bluebook, compiling key resume tips and insights from **300+ successful candidates in IITM**
- Helped organize and fully **verify 400+ student mock resumes** to ensure readiness for internship and placement process

EXTRA-CURRICULARS

National Cadet Corps

- Mentored **NCC** fresher cadets in parade skills, **marching techniques, and discipline, fostering teamwork** & camaraderie
- Participated in the weapon training, Republic Day 2024 and 2023 drill, adventure activities, and a **drug awareness camp**

Amazon ML Summer School

- Selected for the prestigious & competitive Amazon ML Summer School, amongst small number of students nationwide
- Gained comprehensive expertise in **SL, USL, DNNs, RL[19], DR, Seq. Learning**, and Advanced GenAI applications using LLMs

GN6106 - TA

- Assisted as **TA** for **GN6106** & actively **evaluating assignments and mentoring 12** students to improve their performance

Saathi Mentor

- Mentoring **3** freshmen students of my department to help them in facing the challenges of college life through **SAATHI**

1 Data Structures and Algorithm 2. Structured Query Language 3. Object oriented Programming 4. Machine learning 5. Deep Learning 6. Natural Language Processing 7. Computer vision 8. Packet Capture 9. Simple Mail Transfer Protocol 10. JSON Web Token 11. Intersection over Union 12. Optical Character Recognition 13. Bilingual Evaluation Understudy
14. Term Frequency – Inverse Document Frequency 15. Natural Language Toolkit 16. Long Short-Term Memory 17. Root Mean Squared Error 18. Department Mentorship programme 19. Reinforcement Learning, **DNNs** – Deep Neural Networks,